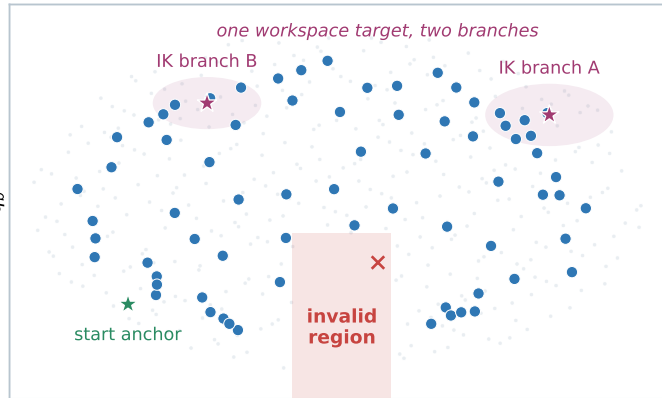


Method-specific node construction under the same 800-node budget

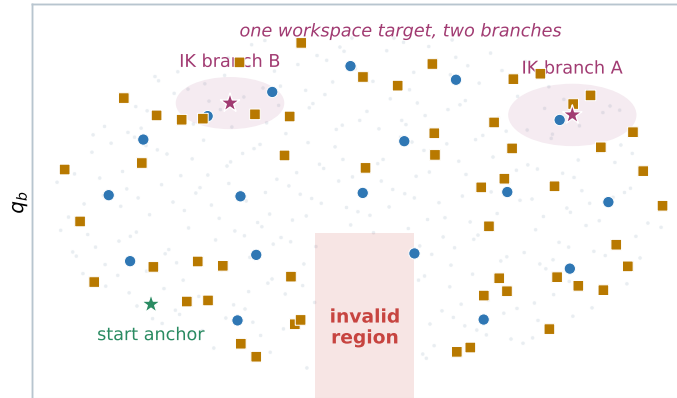
Two-dimensional analogue run with the deployed deterministic GNG and Halton code; marker counts are scaled, labels give the 800-node composition.

GNG-quantized PRM



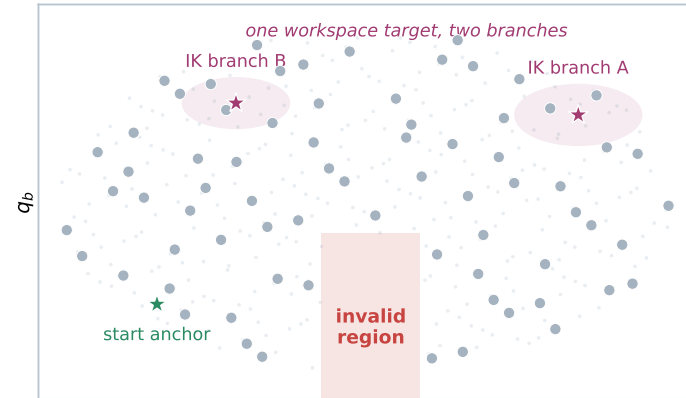
2 anchors + 798 prototypes; learned GNG adjacency is discarded

Guarded GNG



2 anchors + 199 prototypes + 599 low-discrepancy guards

Halton/PRM



2 anchors + 798 direct digit-permuted Halton nodes

● accepted candidate stream

● retained roadmap node

✗ prototype rejected as invalid

■ deterministic guard

All three node sets use the same validated k -nearest-neighbor connection rule, edge limits, and collision predicates.